

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T06 · Percentages

With Answers

Questions with full worked solutions

■ 61 pages

FAMILY MAP

Percentages

F01	Measure or apply a single percentage change	12 Q 46 marks
F02	Recover an original amount	11 Q 34 marks
F03	Model repeated percentage change	3 Q 9 marks
F04	Apply a percentage operator	3 Q 13 marks

- 2 Anna makes cups.
Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.
Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

- (a) Work out the percentage profit Anna makes.
Show your working clearly.

..... %
(4)

Worked Solution 02

Percentage profit

UNIT 2**4 marks****Cost**

$$80 \times 4 \times 6 = 1920 \text{ Swiss francs.}$$

Profit and percentage

$$2160 - 1920 = 240, \quad \frac{240}{1920} \times 100 = 12.5\%.$$

Final answer

$$12.5\%$$

- 5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.
Work out the normal price of the watch.

..... euros

(Total for Question 5 is 3 marks)

Worked Solution 04

Recover a normal price after a percentage reduction

UNIT 2
3 marks**Remaining percentage**

$$100\% - 28\% = 72\%, \quad 0.72N = 198.$$

Solve

$$N = \frac{198}{0.72} = 275.$$

Final answer

275 euros

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%

In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x\%$

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

Worked Solution 08

Successive depreciation followed by growth

UNIT 2
3 marks**Apply the two decreases**

$$750000(1 - 0.04)(1 - 0.065) = 750000(0.96)(0.935) = 673200.$$

Apply the unknown increase

$$673200 \left(1 + \frac{x}{100}\right) = 698445, \quad 1 + \frac{x}{100} = \frac{698445}{673200} = \frac{83}{80} = 1.0375.$$

Find x

$$x = 3.75.$$

Final answer

$$x = 3.75\%$$

- 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.
- Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 7 is 3 marks)

Worked Solution 22

Compound percentage growth

UNIT 2
3 marks**Three annual increases**

$$4000(1.07)^3 = 4900.172 \dots$$

Round

To the nearest Swiss franc: 4900.

Final answer

4900 Swiss francs

Question 03

4MA1-1H Q4

ORIGINAL QUESTION

3 marks

- 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

Worked Solution 03

Reverse percentage

MODULAR UNIT 2

3 marks

Reverse the increase

$$1.15m = 943 \Rightarrow m = 943 \div 1.15 = 820.$$

Final answer

820 km

Question 01

4MA1-1H Q3 demand

ORIGINAL QUESTION UNIT 2

4 marks

3 Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

.....%

Worked solution 01

Percentage profit

MODULAR UNIT 2

4 marks

Demand – Percentage profit

Find revenue

$$1500 + 36(450) = 17700.$$

Find profit

$$17700 - 12500 = 5200.$$

Convert to a percentage

$$\frac{5200}{12500} \times 100 = 41.6\%.$$

Final answer

41.6%

Question 02

4MA1-1H Q6 demand

ORIGINAL QUESTION UNIT 2

4 marks

- 6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket A $\frac{1}{6}$ off normal price

ticket B 20% off normal price

The sale price of ticket **A** is 140 euros.

The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

Worked solution 02

Recover normal ticket prices

MODULAR UNIT 2

4 marks

Demand – Recover normal ticket prices

Ticket A

$$\frac{5}{6}N_A = 140 \Rightarrow N_A = 168.$$

Ticket B

$$0.8N_B = 136 \Rightarrow N_B = 170.$$

Compare

$$170 - 168 = 2.$$

Final answer

2 euros

Question 03

4MA1-1H Q3

ORIGINAL QUESTION

4 marks

- 3 Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.
She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.
Give your answer correct to 3 significant figures.

.....%

Worked Solution 03

Percentage profit

MODULAR UNIT 2

4 marks

Find the income

$$0.70(60)(22) = 924, \quad 0.30(60)(19) = 342.$$

Compare with cost

$$\text{profit} = 924 + 342 - 780 = 486.$$

Convert to a percentage

$$\frac{486}{780} \times 100 = 62.307 \dots \%$$

Final answer

$$62.3\%$$

Original question 06

4MA1-1H Q7 M01

MODULAR UNIT 2

4 marks

- 7 Shop A and Shop B have offers for buying the same type of laptop.

The normal price of the laptop in Shop A is different to the normal price of the laptop in Shop B

Shop A	Shop B
Our normal price £475	Get 15% off our normal price
Get 16% off our normal price	Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

Worked solution 06

Compare two discounts

MODULAR UNIT 2

4 marks

M01 – Compare two discounts**Shop A discount**

$$0.16(475) = 76 \text{ pounds.}$$

Recover Shop B's normal price

$$0.85N_B = 408 \Rightarrow N_B = 480.$$

Shop B discount

$$0.15(480) = 72 \text{ pounds.}$$

Compare

$76 > 72$, so Shop A gives more money off.

Final answer

Shop A (76 pounds versus 72 pounds)

Original question 18

4MA1-2H Q4 M01

MODULAR UNIT 2

2 marks

- 4 N is a number.
17% of N is 357

(a) Work out the value of N

$N = \dots\dots\dots$
(2)

Worked solution 18

Recover a whole from a percentage

MODULAR UNIT 2

2 marks

M01 – Recover a whole from a percentage**Write the percentage equation**

$$0.17N = 357.$$

Solve

$$N = 357/0.17 = 2100.$$

Final answer

2100

Original question 19

4MA1-2H Q4 M02

MODULAR UNIT 2

3 marks

In 2019, the population of a village was 650

In 2020, the population of the village was 806

(b) Work out the percentage increase in the population.

..... %
(3)

Worked solution 19

Find a percentage increase

MODULAR UNIT 2

3 marks

M02 – Find a percentage increase**Find the increase**

$$806 - 650 = 156.$$

Divide by the original

$$156/650 = 0.24.$$

Convert to percent

$$0.24 \times 100 = 24\%.$$

Final answer

24%

Original question 06

4MA1-1H Q6 Q6(a)

ORIGINAL QUESTION**3 marks**

- 6 Asha bought an apartment.

The table gives information about the value of apartments, in euros, and the annual service charge band.

Value (x euros)	Service charge band
$x \geq 700\,000$	A
$600\,000 \leq x < 700\,000$	B
$500\,000 \leq x < 600\,000$	C
$400\,000 \leq x < 500\,000$	D
$0 < x < 400\,000$	E

In 2021, the value of Asha's apartment was 634 400 euros.

The value of Asha's apartment had increased by 4% from its value in 2020

- (a) Has the annual service charge band changed for Asha's apartment?
Show your working clearly.

(3)

Worked solution 06

Chapter: Percentages

MODULAR UNIT 2

3 marks

Q6(a) – Chapter: Percentages

Family: Recover an original amount • Variant: Reverse an increase or changed population/value

Method

A 4% increase has multiplier 1.04.

*Why: Convert the percentage increase to its multiplier.***Method**The 2020 value was $634400/1.04 = 610000$ euros.*Why: Reverse the increase by dividing the later value by the multiplier.***Method**

Both 610000 euros and 634400 euros are in band B, so the band did not change.

*Why: Compare each value with the stated band boundaries.***Final answer****decision:** No; the 2020 value was 610000 euros and both years are band B

Original question 12

4MA1-1H Q13 Q13

ORIGINAL QUESTION**5 marks****13** Antoine is going on holiday.

He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{3}{8}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$406

Work out how much Antoine has to pay for Payment 2

\$

Worked solution 12

Chapter: Percentages

MODULAR UNIT 2

5 marks

Q13 – Chapter: Percentages

Family: Apply a percentage operator • Variant: Find a percentage of an amount

MethodPayments 1 and 2 are $\frac{3}{8} + 45\% = 0.375 + 0.45 = 0.825$ of the total.*Why: Convert the fraction and percentage to the same form before adding.***Method**Payment 3 is the remaining $1 - 0.825 = 0.175$ of the total.*Why: All three payments together make the full cost.***Method**If the total cost is T, then $0.175T = 406$.*Why: Match the known third payment to its fraction of the total.***Method** $T = 406 / 0.175 = 2320$ dollars.*Why: Divide by the remaining proportion to find the whole.***Method**Payment 2 is $0.45 \times 2320 = 1044$ dollars.*Why: Apply the given 45% to the total cost.***Final answer****Payment 2: \$1044**

Original question 22

4MA1-2H Q8 Q8

ORIGINAL QUESTION**5 marks**

- 8 Behnaz makes 300 celebration cards so that

number of birthday cards : number of anniversary cards : number of congratulations cards = 7:5:3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.

Give your answer in its simplest form.

Worked solution 22

Chapter: Percentages

MODULAR UNIT 2

5 marks

Q8 – Chapter: Percentages

Family: Apply a percentage operator • Variant: Combine percentage with fraction or ratio groups

MethodThere are $7 + 5 + 3 = 15$ ratio parts, so one part represents $300/15 = 20$ cards.*Why: Convert the ratio to actual card counts.***Method**There are $7 \times 20 = 140$ birthday cards, and $(2/5) \times 140 = 56$ have numbers.*Why: Apply the stated fraction to the birthday cards.***Method**There are $5 \times 20 = 100$ anniversary cards, and $36\% \times 100 = 36$ have numbers.*Why: Apply the stated percentage to the anniversary cards.***Method**The total with numbers is $56 + 36 = 92$, so the fraction is $92/300 = 23/75$.*Why: Combine the two groups and simplify the fraction.***Final answer**fraction: $23/75$

Original question 03

4MA1-1H Q4

ORIGINAL QUESTION**5 marks**

- 4 Sarah makes and sells mugs.

One day she makes 150 mugs.

Her total cost for making these mugs is £1140

Of these mugs

$\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	£8.50
Medium	£11.20
Large	£14.20

Sarah sells all 150 mugs.

Work out her percentage profit.

Give your answer correct to the nearest whole number.

.....%

Worked solution 03

Chapter: Percentages

MODULAR UNIT 2

5 marks

Q4 – Chapter: Percentages

Family: Measure or apply a single percentage change • Variant: Profit, loss or percentage margin from costs and revenue

Method

Small mugs = $(2/5) \times 150 = 60$ and medium mugs = $0.32 \times 150 = 48$.

Why: Convert the fraction and percentage of 150 mugs.

Method

Large mugs = $150 - 60 - 48 = 42$.

Why: The remaining mugs are large.

Method

Income = $60 \times 8.50 + 48 \times 11.20 + 42 \times 14.20 = 1644$ pounds.

Why: Multiply each quantity by its selling price and add.

Method

Profit = $1644 - 1140 = 504$ pounds.

Why: Subtract total cost from total income.

Method

Percentage profit = $(504/1140) \times 100 = 44.2105\ldots\% \approx 44\%$.

Why: Use profit divided by cost and round to the nearest whole percent.

Final answer

percentage profit: 44%

Original question 21

4MA1-2H Q8 Q8(a)

ORIGINAL QUESTION**2 marks**

- 8 The table gives information about the average house price in England in 2018 and in 2019

Year	2017	2018	2019
Average house price (£)		228 314	231 776

- (a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

..... %
(2)

Worked solution 21

Chapter: Percentages

MODULAR UNIT 2

2 marks

Q8(a) – Chapter: Percentages

Family: Measure or apply a single percentage change • Variant: Calculate a percentage increase or decrease

MethodPercentage increase = $(231776 - 228314) / 228314 \times 100$.*Why: Use change divided by original value.***Method** $= 1.516\ldots\% \approx 1.5\%$.*Why: Round to one decimal place.***Final answer**

percentage increase: 1.5%

Original question 22

4MA1-2H Q8 Q8(b)

ORIGINAL QUESTION**3 marks**

The average house price in 2019 was 7.7% greater than the average house price in 2017

- (b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£
(3)

Worked solution 22

Chapter: Percentages

MODULAR UNIT 2

3 marks

Q8(b) – Chapter: Percentages

Family: Recover an original amount • Variant: Reverse an increase or changed population/value

Method2019 price = 1.077×2017 price.*Why: A 7.7% increase has multiplier 1.077.***Method**2017 price = $231776 / 1.077 = 215205.19...$ pounds.*Why: Reverse the percentage increase.***Method**

To 3 significant figures, the 2017 price is £215000.

*Why: Round as requested.***Final answer****2017 price: £215000**

- 8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says,

“The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017”

- (a) Is Brigitte correct?

You must show how you get your answer.

(4)

Worked solution

January 2020 | Question 8 | 4 marks

Family: Measure or apply a single percentage change • Variant: Calculate a percentage increase or decrease

Method / working

1. 2016->2017 increase= $45/500 \times 100=9\%$.
2. 2017->2018 increase= $47/545 \times 100\sim 8.62\%$.
3. Compare the two percentages.
4. Therefore Brigitte is not correct.

Final answer: No: 9% is greater than 8.6%

Henri wants to insure his car.

He gets a discount of 15% off the normal price.

Henri pays 952 euros for his car insurance after the discount.

(b) Work out the discount that Henri gets.

..... euros
(3)

Worked solution

January 2020 | Question 8 | 3 marks

Family: Recover an original amount • Variant: Reverse a decrease or sale discount

Method / working

1. After 15% discount, 952 is 85% of the original.
2. $\text{Original} = 952 / 0.85 = 1120$ euros.
3. $\text{Discount} = 1120 - 952 = 168$ euros.

Final answer: 168 euros

3 Gladys buys a table for \$465 to sell in her shop.

She sells the table for \$520

- (a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

..... %
(3)

Worked solution

January 2021 | Question 3 | 3 marks

Family: Measure or apply a single percentage change • Variant: Profit, loss or percentage margin from costs and revenue

Method / working

1. Profit = $520 - 465 = 55$ dollars.
 2. Percentage profit = $(55/465) \times 100 = 11.8279\ldots\%$.
 3. To 3 s.f., the percentage profit is 11.8%.
- Final answer: 11.8%

Gladys has a sale in her shop.

She decreases all the normal prices by 12%

The normal price of an armchair was \$550

(b) Work out the sale price of the armchair.

\$.....
(3)

Worked solution

January 2021 | Question 3 | 3 marks

Family: Measure or apply a single percentage change • Variant: Apply a stated increase, decrease or discount

Method / working

1. A 12% reduction leaves 88%.
 2. Sale price = 0.88×550 .
 3. Therefore the sale price is \$484.
- Final answer: \$484

6 Mariana sells bags of bird food.

The bags that Mariana sold last week each contained 12 kg of seeds.

The bags that she is going to sell next week will each contain a mixture of nuts and seeds where for each bag

$$\text{weight of nuts : weight of seeds} = 4 : 5$$

The total weight of the nuts and the seeds in each bag will be 19.35 kg

The weight of seeds in each bag that Mariana sells next week will be less than the weight of seeds in each bag that Mariana sold last week.

Work out this decrease as a percentage of the weight of seeds in each bag that Mariana sold last week.

Give your answer correct to one decimal place.

..... %

Worked solution

June 2021 | Question 6 | 4 marks

Family: Measure or apply a single percentage change • Variant: Calculate a percentage increase or decrease

Method / working

1. Total seed mass is 19.35 kg and the ratio has 9 parts, so one part is 2.15 kg.
2. The original seed mass is $5 \times 2.15 = 10.75$ kg.
3. Percentage decrease = $(10.75 - 9.5) / 10.75 \times 100$.
4. Evaluate and round: 10.4%.

Final answer: 10.4%

11 Chengbo sold a house for 180 000 yuan.

The amount for which he sold the house is 24% more than the amount he paid for the house.

- (a) Work out how much Chengbo paid for the house.
Give your answer correct to 3 significant figures.

..... yuan
(3)

Worked solution

June 2021 | Question 11 | 3 marks

Family: Recover an original amount • Variant: Reverse an increase or changed population/value

Method / working

1. Selling price is 24% above purchase price, so purchase price $= 180000 / 1.24$.
2. Evaluate $180000 / 1.24$.
3. State 145000 yuan.

Final answer: 145000 yuan

Zhi bought a house on 1st January 2017
When she bought the house, its value was 120 000 yuan.

The value of the house increased by 1.8% per year.

- (b) Work out the value of Zhi's house on 1st January 2020
Give your answer correct to 3 significant figures.

..... yuan
(3)

Worked solution

June 2021 | Question 11 | 3 marks

Family: Model repeated percentage change • Variant: Repeated growth or decay with known rate and time

Method / working

1. The value grows for three years from 1 January 2017 to the stated date: 120000×1.018^3 .
2. Evaluate to 126597.34... yuan.
3. Round to the nearest thousand as required: 127000 yuan.

Final answer: 127000 yuan

- 8 In 2018, the population of Sydney was 5.48 million.
This was 22% of the total population of Australia.

Work out the total population of Australia in 2018
Give your answer correct to 3 significant figures.

..... million

Worked solution

November 2021 | Question 8 | 3 marks

Family: Recover an original amount | Variant: Reverse an increase or changed population/value

METHOD / WORKING

1. Sydney's population was 22% of Australia's total population.

2. $0.22 \times \text{total} = 5.48$ 3. $\text{total} = \frac{5.48}{0.22} = 24.909\ldots$

4. Correct to 3 significant figures,

5. $24.909\ldots \approx 24.9$

6. Answer: 24.9 million

FINAL ANSWER

24.9 million

- 6 Victor buys 12 bottles of apple juice for a total cost of \$21
Victor sells all 12 bottles at \$2.45 each bottle.

Work out Victor's percentage profit.

..... %

Worked solution

November 2021 | Question 6 | 3 marks

Family: Measure or apply a single percentage change | Variant: Profit, loss or percentage margin from costs and revenue

METHOD / WORKING

1. Total selling price:

$$2. \quad 12 \times 2.45 = 29.40$$

3. Profit:

$$4. \quad 29.40 - 21 = 8.40$$

5. Percentage profit:

$$6. \quad \frac{8.40}{21} \times 100 = 40$$

7. Answer: 40%

FINAL ANSWER

40%

- 9 In a sale, normal prices are reduced by 20%

A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

..... rupees

(Total for Question 9 is 3 marks)

Answer reference | May-Nov 2020 | Question 9

Family: Apply a percentage operator • Variant: Combine percentage with fraction or ratio groups

9	E.g. $1 - 0.2 (= 0.8)$ or $100(\%) - 20(\%) (= 80(\%))$ or $\frac{1080}{80} (= 13.5)$ oe		3	M1
	E.g. $1080 \div 0.8$ or $1080 \div 80 \times 100$ or ' 13.5 ' $\times 100$ $1080 \times 100 \div 80$			M1 for a complete method
		1350		A1
				Total 3 marks

Worked solution

May-Nov 2020 | Question 9 | 3 marks

Family: Apply a percentage operator • Variant: Combine percentage with fraction or ratio groups

Method

A reduction of 20\% means the sale price is 80\% of the normal price.

$$0.80 \times \text{normal price} = 1080$$

$$\text{normal price} = (1080) / (0.80) = 1350$$

Answer: 1350 rupees

6 Shane bought a car.

The amount Shane paid for the car was \$32 000

Theresa also bought a car.

To pay for this car, Theresa paid a deposit of \$18 000 together with 14 monthly payments of \$1160

Theresa paid more for her car than Shane paid for his car.

(a) Work out how much more Theresa paid as a percentage of the amount Shane paid.

..... %
(4)

Kylie bought a van.

After 1 year, the value of the van was \$39 865

During this year, the value of the van decreased by 15%

(b) Work out the value of the van when Kylie bought it.

\$
(3)

(Total for Question 6 is 7 marks)

Answer reference | January 2023 | Question 6

Family: Measure or apply a single percentage change • Variant: Calculate a percentage increase or decrease

6	(a)	$18\,000 + 14 \times 1160 (= 34\,240)$ oe or $18\,000 + 16\,240 (= 34\,240)$		4	M1
		$"34\,240" - 32\,000 (= 2240)$ or $\frac{"34\,240"}{32\,000} (= 1.07)$			M1
		$\frac{"2240"}{32\,000} (\times 100)$ or $\frac{"34\,240"}{32\,000} \times 100 (= 107)$ or $"1.07" - 1 (= 0.07)$			M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	7		A1
	(b)	e.g. $1 - 0.15 (= 0.85)$ or $100(\%) - 15(\%) (= 85(\%))$		3	M1
		e.g. $39\,865 \div 0.85$ or $39\,865 \div 85 \times 100$ oe			M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	46 900		A1
					Total 7 marks

Worked solution

January 2023 | Question 6 | 7 marks

Family: Measure or apply a single percentage change • Variant: Calculate a percentage increase or decrease

Official final mark-scheme pages are embedded immediately after this question.

- 9 Hari scored 140 points playing a game on Tuesday.
This was 12% more points than he scored playing the game on Monday.
Work out the number of points that Hari scored playing the game on Monday.

(Total for Question 9 is 3 marks)

Worked solution

November 2025 | Question 9 | 3 marks

Family: Recover an original amount • Variant: Reverse an increase or changed population/value

Method

Tuesday's score was 12% more than Monday's score, so Tuesday was 112% of Monday.

$$1.12 \times \text{Monday} = 140$$

$$\text{Monday} = (140) / (1.12) = 125$$

Answer: 125 points